

Program : Diploma in Artificial Intelligence & Machine Learning	
Course Code : 6341C	Course Title: Computer Vision and Image Processing
Semester : 6	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- Give knowledge about the basic theory and algorithms related to Digital Image Processing..
- To develop skills in designing and implementing basic computer vision applications.
- To understand how to capture and extract relevant visual information from images and videos to automatically interpret the 2D/3D scene.

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of AI		Introduction to Artificial Intelligence	III
Basic of ML		Introduction to Machine Learning	IV

Course Outcomes :

On completion of the course, the student will be able to:

CO_n	Description	Duration (Hours)	Cognitive Level
CO1	Explain the fundamental concepts of Image processing.	12	Understanding
CO2	Summarise the phases of image processing and its techniques.	15	Understanding
CO3	Explain the fundamentals of Computer vision	16	Understanding
CO4	Outline the techniques to perceive 3D from 2D images	15	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Descriptions	Duration (Hours)	Cognitive Level
CO1	Explain the fundamental concepts of Image processing		
M1.01	Describe the Elements of Image Processing.	1	Understanding
M1.02	Describe Color processing	1	Understanding
M1.03	Describe various Image Processing Techniques	2	Understanding
M1.04	Summarize Digital Image fundamentals	4	Understanding
M1.05	Classify Digital Images	1	Understanding
M1.06	Outline the applications of Image Processing	3	Understanding
Contents : Concepts: Simple image model, Fundamental steps in image processing, Elements of Image Processing-Color processing-basics of color, color models in images/video, Image processing techniques Digital Image fundamentals –elements of visual perception, light and electromagnetic spectrum, Image sensing and Acquisition, sampling and quantization ,Relationship between pixels, types of image., Application of image processing			
CO2	Summarise the phases of image processing and its techniques.		
M2.01	Explain different Intensity Transformation	2	Understanding
M2.02	Explain spatial filtering methods	2	Understanding
M2.03	Describe Image degradation/Restoration process model	2	Understanding
M2.04	Illustrate Color image processing	3	Understanding
M2.05	Describe types of Image segmentation	3	Understanding
M2.06	Explain Image compression methods	3	Understanding
	Series Test -1	1	
Contents: Intensity Transformation –image negatives, Log transformation and power Law(Gamma)transformation, Histogram processing. spatial filtering methods- Smoothing Spatial Filters, Sharpening Spatial Filters. Image degradation/Restoration process model- Noise probability density functions. Color image processing- Color fundamentals, Color models – RGB, CMYK, HSI. Full color image processing. Image segmentation- Region-based segmentation, Segmentation using Morphological Watersheds. Image compression methods- Huffman coding,LZW coding			
CO3	Explain the fundamentals of Computer vision		

M3.01	Imaging and Image Representation	4	Understanding
M3.02	Binary Image Analysis	4	Understanding
M3.03	Introduction to pattern recognition concepts	4	Understanding
M3.04	Describe Filtering and enhancing images	4	Understanding
Contents: Imaging and Image Representation- Imaging Devices, 3D structure from 2D images, Five frames of reference. Binary Image Analysis- Pixels and Neighborhoods, Applying masks to images, Counting the objects in an image, Connected components labelling. Introduction to pattern recognition concepts used for representation, classifiers, and decision trees. Filtering and enhancing images - Image smoothing, median filtering and Gaussian filtering.			
CO4	Outline the techniques to perceive 3D from 2D images		
M4.01	Illustrate Color and shading	3	Understanding
M4.02	Explain Content-based image retrieval	3	Understanding
M4.03	Demonstrate Motion from 2D image sequences	3	Understanding
M4.04	Explain Matching in 2D	2	Understanding
M4.05	Perceiving 3D from 2D images	4	Understanding
	Series Test -2	1	
Contents: Color and Shading- Color bases, Color histograms, Color segmentation, Shading. Content based image retrieval- Image distance measures, Database organization. Motion from 2D image sequences- : Computing Motion Vectors, Computing paths of moving points, Detecting significant changes in video. Overview of image segmentation. Matching in 2D- Registration of 2D data, Representation of points, Affine mapping functions, 2D object recognition via Affine and Relational Matching. Perceiving 3D from 2D images- Labeling of line drawings from blocks world, 3D cues available in 2D images, Perspective imaging model, Depth perception from stereo, 3D Affine transformations, 3D object reconstruction.			

Text / Reference

T/R	Book Title/Author
T1	Linda G. Shapiro, George C. Stockman, "Computer Vision", Prentice Hall, 1st Ed., 2001.
R1	David A. Forsyth, Jean Ponce, "Computer Vision: A Modern Approach", 2nd Ed., 2011.
R2	Richard Szeliski, "Computer Vision: Algorithms and Applications", Springer, 1st Ed., 2010.

Online Resources

Sl.No	Website Link
1	https://www.v7labs.com/
2	https://www.geeksforgeeks.org/